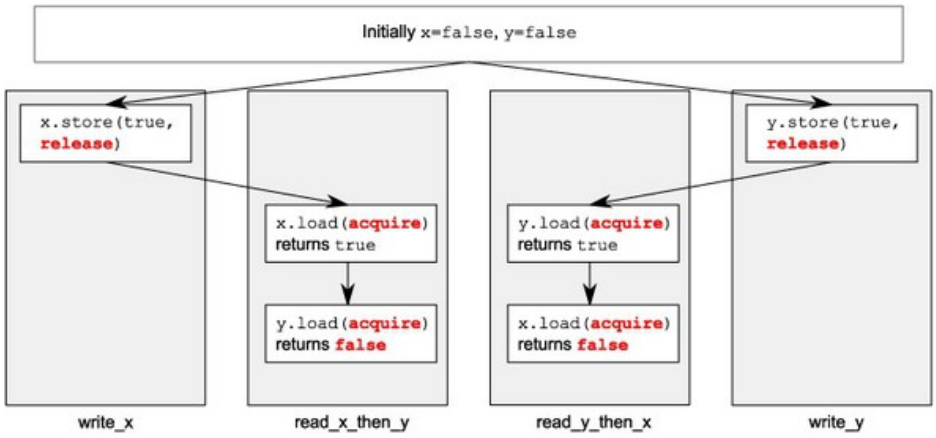
Figure 5.6 shows the happens-before relationships from listing 5.7, along with a possible outcome where the two reading threads each have a different view of the world. This is possible because there's no happens-before relationship to force an ordering, as described previously.

Figure 5.6. Acquire-release and happens-before



In order to see the benefit of acquire-release ordering, you need to consider two stores from the same thread, like in listing 5.5. If you change the store to `y` to use `memory_order_release` and the load from `y` to use `memory_order_acquire` like in the following listing, then you actually impose an ordering on the operations on `x`.

Listing 5.8. Acquire-release operations can impose ordering on relaxed operations